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BITS Pilani
Pilani Campus

Course	Applied Data Science (S1-25_MBACCZG581)
EC1: 3 Phase Mini	Project Plan
Presented by	Group 11
Title	Factors Influencing Housing Prices – Boston (EDA)
Date	14-02-2026

Group 11 Team Introduction

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ZG581: APPLIED DATA SCIENCES - MINI PROJECT ALLOTMENT TABLE

EC1 PROJECT ALLOCATION SHEET

Complete By: 3rd Dec 2025 23:59 hrs IST

Instructions:

Total:	35

PR. ID	PROJECT TITLE	Enter your group number in the box to select the project.				
		First Group	# in group	Second Group	# in group	Total participants
P13	**Housing Price Prediction** – Boston Housing dataset (UCI ML).	11	4	select		4

Factors Influencing Housing Prices – Boston (EDA)

Acknowledgement

- We sincerely thank BITS Pilani Faculty (Shree Gokhale Jayanti) for providing this opportunity to apply data science concepts to real-world problems.
- We also acknowledge the support of our peers and the availability of public datasets from UCI Machine Learning Repository and Kaggle, which made this analysis possible.

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Factors Influencing Housing Prices – Boston (EDA)

Problem & Objective

- Understand which property/neighborhood factors align with higher/lower prices.
- Exploratory only; no predictive modelling.

Notes: The aim is clear insights for decisions across real-estate, policy, and buyers.

Dataset & Method

- Boston Housing (UCI via Kaggle) | 506 × 14 | CHAS categorical | no missing values.
- Cleaning: headers, types, IQR capping; EDA with scatter plots, histograms, heatmap.

Notes: We focused on readable charts with business meaning per EC1's guidance.

Reference : Please refer phase 1 report for more details for problem statement and dataset use case..

Factors Influencing Housing Prices – Boston (EDA)

Insight #1: Rooms (RM)

- More rooms → higher price
- KPI: RM is a valuation driver.

Notes: Emphasize RM in listings and pricing

Insight #2: Crime (CRIM)

- Higher crime → lower price
- KPI: neighborhood risk score

Notes: Safer areas retain premium pricing

Reference : Please refer phase 2 report (created on Jupyter Note book) for more details of insights.

Factors Influencing Housing Prices – Boston (EDA)

Insight #3: School (PTRATIO)

- Lower PTRATIO → higher price
- KPI: school advantage

Notes: Families value better student attention

Heatmap Summary

- Drivers: RM (+), LSTAT (–), CRIM (–), PTRATIO (–)

Notes: These four explain most of what we see in pricing patterns

Reference : Please refer phase 2 report (created on Jupyter Note book) for more details of insights.

Factors Influencing Housing Prices – Boston (EDA)

Recommendations

- Developers: highlight room count, school edge, and safety
- Policy: invest in safety and education.
- Buyers: check RM↑, PTRATIO↓, CRIM↓, LSTAT↓

Notes: These are practical and easy to execute.

Reference : Please refer phase 3 report (on Jupyter Note book) for more details of insights.

Factors Influencing Housing Prices – Boston (EDA)

Ethics & Limits

- Sensitive attributes (LSTAT) handled carefully.
- MEDV is capped at 50; luxury analysis is limited.
- IQR capping clarifies visuals but compresses extremes.

Notes: We followed responsible data-use and transparency principles**.**

Reference : Please refer phase 3 report (.pdf file) for more details of Ethics and Limits.

Thank You